



## Lab 06: Departmental Subnetting

IT · Finance · HR PCs · Switch3 (2960) · Router0 (2911)

Lab #06

Topic: Subnetting & VLANs

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

### 1. Lab Objectives

- Create three VLANs on Switch3 for IT, Finance and HR departments.
- Connect all department PCs to Switch3 on Fa0/1, Fa0/2 and Fa0/3.
- Uplink Switch3 to Router0 (2911) on Fa0/4 → Gig0/0 as a trunk.
- Configure Router-on-a-Stick sub-interfaces for inter-department routing.
- Verify IT, Finance and HR can all reach each other through Router0.

### 2. Network Topology

Switch3 (2960-24TT) is central. IT PC on Fa0/1, Finance PC on Fa0/2, HR PC on Fa0/3. Switch3 uplinks to Router0 (2911) on Fa0/4 → Gig0/0 (trunk link). Router0 uses sub-interfaces to route between the three department VLANs.

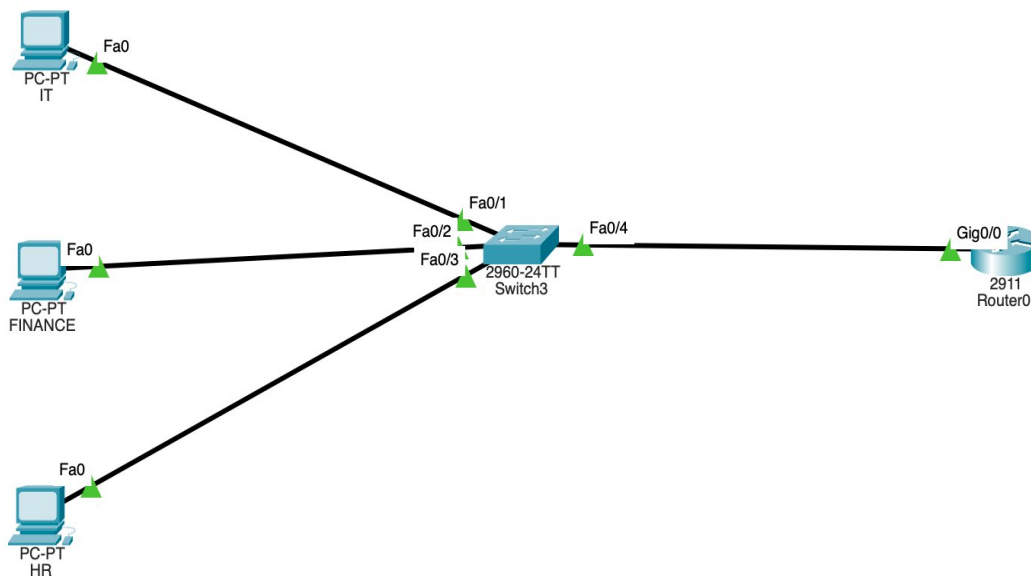


Figure 1 – IT, Finance, HR PCs on Switch3 (2960) connected to Router0 (2911)

| Department | PC Name | Switch Port | VLAN    | IP Address    | Gateway      |
|------------|---------|-------------|---------|---------------|--------------|
| IT         | IT      | Fa0/1       | VLAN 10 | 192.168.10.10 | 192.168.10.1 |
| Finance    | FINANCE | Fa0/2       | VLAN 20 | 192.168.20.10 | 192.168.20.1 |



|    |    |       |         |               |              |
|----|----|-------|---------|---------------|--------------|
| HR | HR | Fa0/3 | VLAN 30 | 192.168.30.10 | 192.168.30.1 |
|----|----|-------|---------|---------------|--------------|

## 3. Configuration

### 3.1 Switch3 – VLANs, Access Ports, Trunk

```
Switch3(config)#vlan 10
Switch3(config-vlan)#name IT
Switch3(config-vlan)#exit
Switch3(config)#vlan 20
Switch3(config-vlan)#name FINANCE
Switch3(config-vlan)#exit
Switch3(config)#vlan 30
Switch3(config-vlan)#name HR
Switch3(config-vlan)#exit
Switch3(config)#interface Fa0/1
Switch3(config-if)#switchport mode access
Switch3(config-if)#switchport access vlan 10
Switch3(config-if)#exit
Switch3(config)#interface Fa0/2
Switch3(config-if)#switchport mode access
Switch3(config-if)#switchport access vlan 20
Switch3(config-if)#exit
Switch3(config)#interface Fa0/3
Switch3(config-if)#switchport mode access
Switch3(config-if)#switchport access vlan 30
Switch3(config-if)#exit
Switch3(config)#interface Fa0/4
Switch3(config-if)#switchport mode trunk
Switch3(config-if)#exit
Switch3#write memory
```

### 3.2 Router0 – Sub-interfaces

```
Router0(config)#interface GigabitEthernet0/0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#interface GigabitEthernet0/0.10
Router0(config-subif)#encapsulation dot1Q 10
Router0(config-subif)#ip address 192.168.10.1 255.255.255.0
Router0(config-subif)#exit
Router0(config)#interface GigabitEthernet0/0.20
```



```
Router0(config-subif)#encapsulation dot1Q 20
Router0(config-subif)#ip address 192.168.20.1 255.255.255.0
Router0(config-subif)#exit
Router0(config)#interface GigabitEthernet0/0.30
Router0(config-subif)#encapsulation dot1Q 30
Router0(config-subif)#ip address 192.168.30.1 255.255.255.0
Router0(config-subif)#exit
Router0#write memory
```

---

## 4. Verification

```
IT>ping 192.168.20.10
IT>ping 192.168.30.10
Switch3#show vlan brief
Switch3#show interfaces trunk
```

---

## 5. Key Takeaways

- VLANs isolate department traffic at Layer 2 — improving security and performance.
- Router-on-a-Stick uses ONE trunk link with dot1Q sub-interfaces to route all VLANs.
- The trunk port must be configured first — sub-interfaces fail if the trunk is missing.
- Each department PC uses its sub-interface IP as its default gateway.
- In production, a Layer 3 switch handles inter-VLAN routing faster in hardware.